Lab 2.2 Measuring the Z0 of PCB Traces

Leo Marcinkus

Ruban Ford Alfred

Alex Martinez

System overview

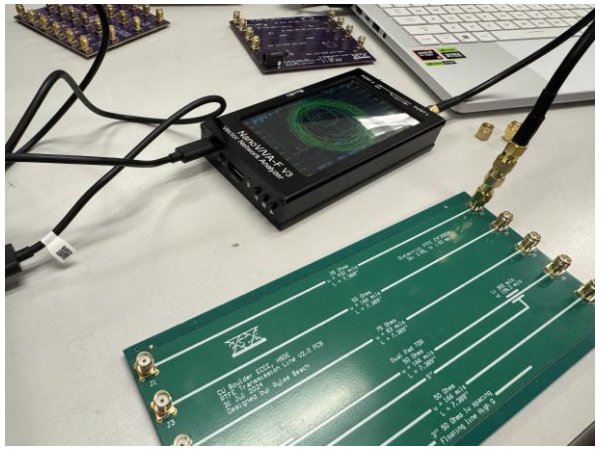
Question: What is needed to correctly understand trace impedance? Do online calculators and 2D field solvers predict the ZO we measure, and how well do they agree with each other?

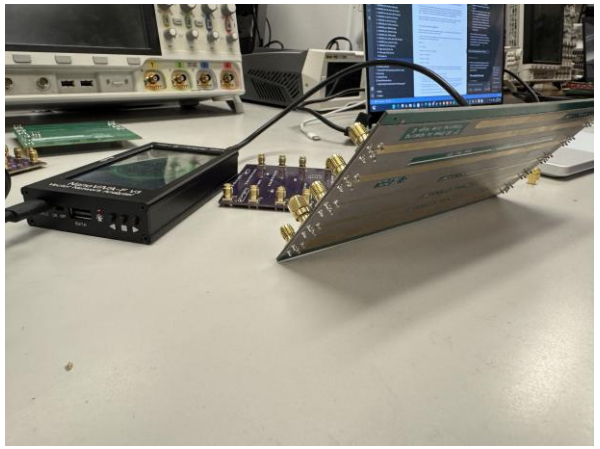
DUT: CU HSDE PTFE Transmission Line V2.0 Board

Same nominal inputs in every tool: Dk = 2.55, H = 1.52 mm (59.842 mils), t = 1.4 mil

Instrument: Calibrated 1-port VNA, 801 points (start = step), .s1p files into an ADS virtual TDR

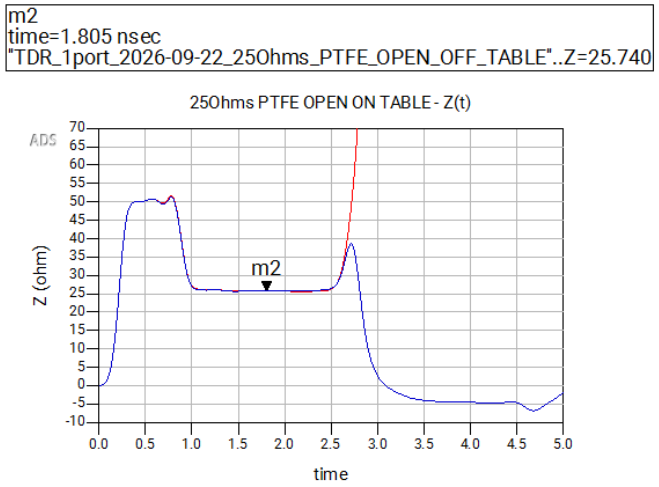
Conditions: Far end **open** and **shorted**. Board flat on the table (**traces down**) and on its edge (**traces in air**)

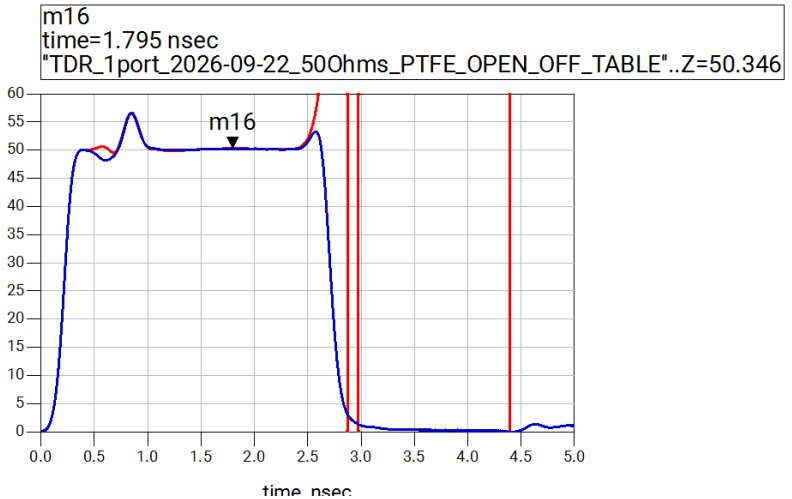


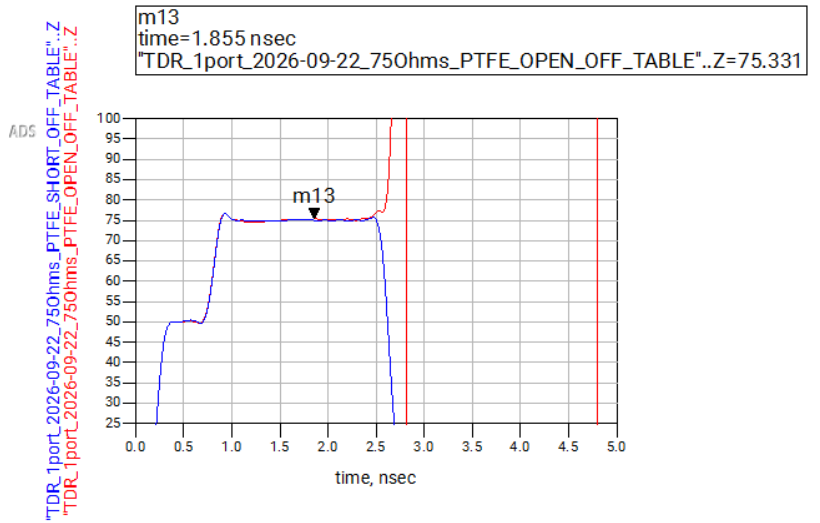




Results from our TDR







Z_measured *= 25.740 ohms*

Z_measured *= 50.346 ohms*

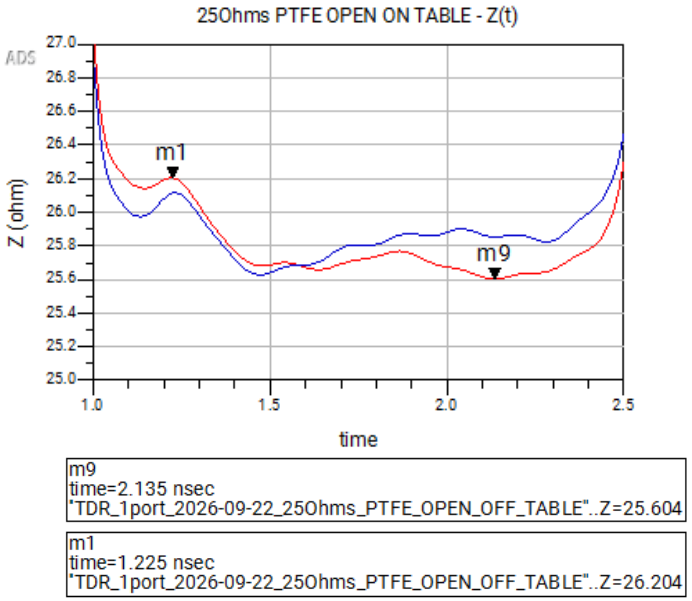
Z_measured *= 75.331 ohms*

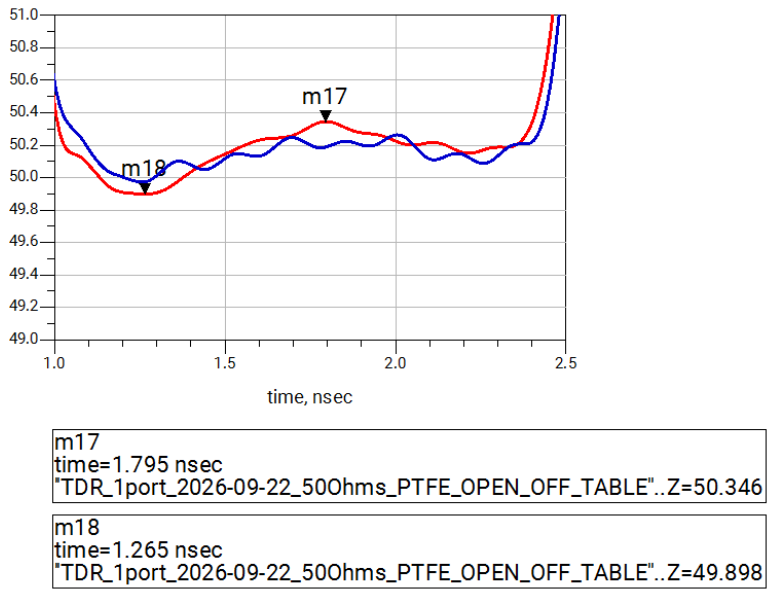
Red → Open at the end

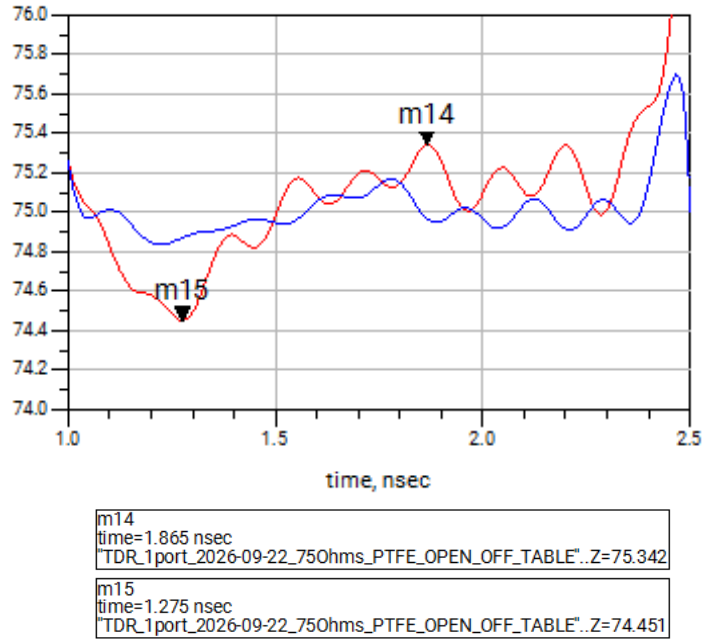
Blue → Shorted at the end

The measured value is pretty close to our nominal values. This was what we were expecting to see!

Results from our TDR - cont







Z_max *= 26.204 ohms*
Z_min *= 25.604 ohms*
Z_range *= 25.904 +/- 0.3 ohms*

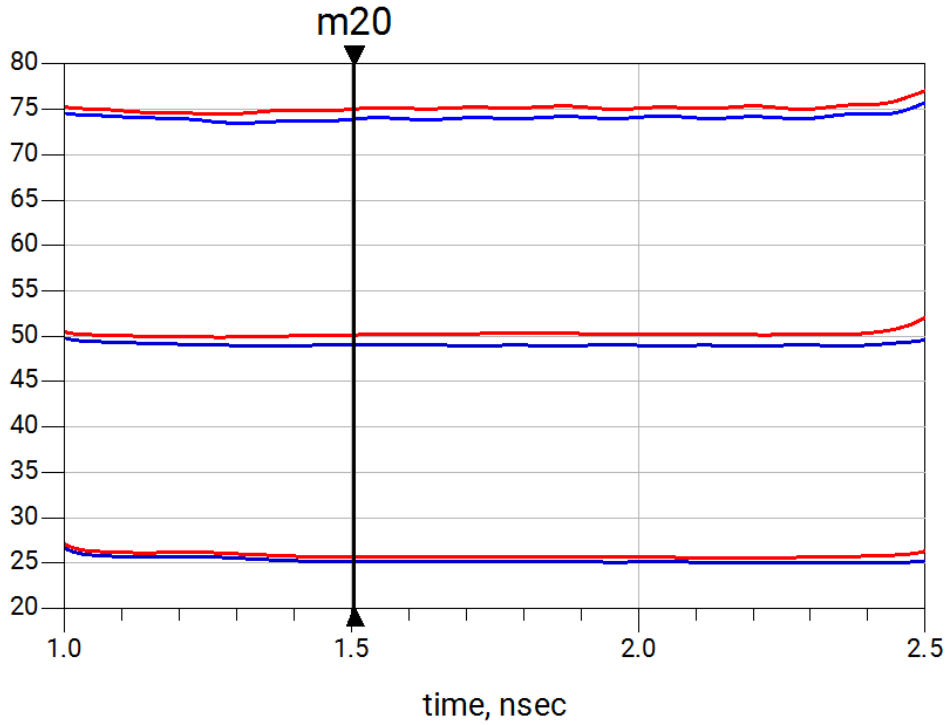
Z_max *= 50.346 ohms*
Z_min *= 49.898 ohms*
Z_range *= 50.122 +/- 0.2 ohms*

Z_max *= 75.342 ohms*
Z_min *= 74.451 ohms*
Z_range *= 74.896 +/- 0.4 ohms*

Red → Open at the end
Blue → Shorted at the end

Every measurement has some uncertainty, so we report each impedance as a nominal value ± its uncertainty. Across all three lines, our uncertainty is about ±0.2–0.4 Ω, roughly 1% of the nominal impedance.

Table is part of the DUT!



m20
time=1.505 nsec
"TDR_1port_2026-09-22_75Ohms_PTFE_OPEN_OFF_TABLE"..Z=75.015
"TDR_1port_2026-09-22_75Ohms_PTFE_OPEN_ON_TABLE"..Z=73.926
"TDR_1port_2026-09-22_50Ohms_PTFE_OPEN_OFF_TABLE"..Z=50.149
"TDR_1port_2026-09-22_50Ohms_PTFE_OPEN_ON_TABLE"..Z=49.061
"TDR_1port_2026-09-22_25Ohms_PTFE_OPEN_OFF_TABLE"..Z=25.690
"TDR_1port_2026-09-22_25Ohms_PTFE_OPEN_ON_TABLE"..Z=25.187

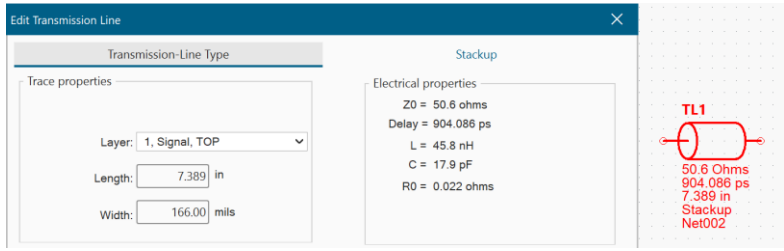
Red → Off the table
Blue → On the table

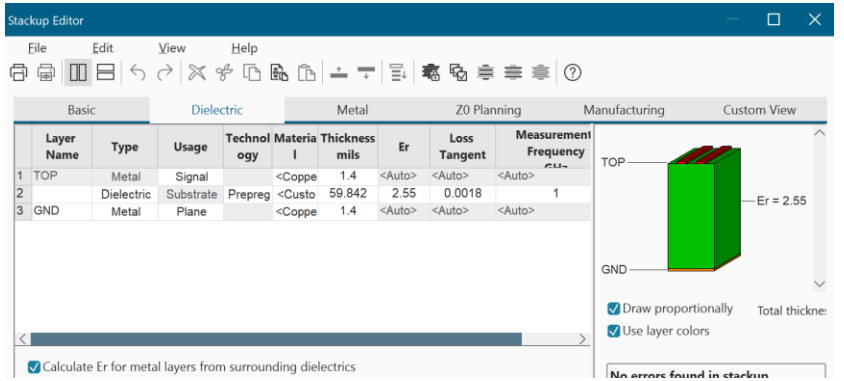
2 Cases was done where the trace is measured
i) Facing away from the Table (off the table)
ii) Facing down on the Table (on the table)

The measurement environment matters: facing the trace down on the table replaces air (Dk = 1) above the microstrip with the table material (Dk > 1), which raises the effective Dk and lowers Z0 by up to about 1.1 Ω, 2-4× our measurement uncertainty.

Approximations vs 2D Field solvers

Approximations	25ohms TW=430mils	50ohms TW=166mils	75ohms TW=83mils
All about circuits	24.8	49.6	74.8
Saturn	**25.8**	**51.2**	**76.4**
Elektroda	24.8	49.6	74.9
Approximations: mean (spread)	25.13 Ω (1.0 Ω, 4.0%)	50.13 Ω (1.6 Ω, 3.2%)	75.37 Ω (1.6 Ω, 2.1%)
2D Field solver	**25ohms TW=430mils**	**50ohms TW=166mils**	**75ohms TW=83mils**
HyperLynx	25.4	50.6	75.8
Sierra	**24.99**	**50.02**	**75.1**
Ansys Q2D	24.78	49.36	74.73
2D field solvers: mean (spread)	25.06 Ω (0.62 Ω, 2.5%)	49.99 Ω (1.24 Ω, 2.5%)	75.21 Ω (1.07 Ω, 1.4%)
Measured	***25.904 +/- 0.3 ohms***	***50.122 +/- 0.2 ohms***	***74.896 +/- 0.4 ohms***





Uncertainty in input parameters

Sensitivity

Input	ΔX (1%)	Z0 at +1% / −1%	ΔZ0 for a 1% change	Sensitivity dZ0/dX	Relative sensitivity (%Z0 per 1% of input)
Trace width, w	1.66 mil	49.689 / 50.340	−0.326 Ω	−0.196 Ω/mil	-0.65
Trace thickness, t	0.014 mil	50.009 / 50.015	−0.003 Ω	−0.25 Ω/mil	-0.007
Dielectric thickness, h	0.0152 mm (0.598 mil)	50.340 / 49.680	**+0.330 Ω**	+0.55 Ω/mil (21.7 Ω/mm)	**+0.66**
Dk	0.0255	49.800 / 50.227	−0.213 Ω	−8.37 Ω per unit Dk	-0.43

Sensitivity * Manufacturing tolerance

ΔW (20%)	**ΔH(+/- 2mil)**	**ΔT(15%)**	**ΔDk(+/-0.04)**	**ΔZ0**
+/- 6.51Ω	+/- 1.10Ω	+/- 0.05Ω	+/- 0.33Ω	+/- 8Ω
ΔW (5%)	**ΔH(+/- 2mil)**	**ΔT(15%)**	**ΔDk(+/-0.04)**	**ΔZ0**
+/- 1.63Ω	+/- 1.10Ω	+/- 0.05Ω	+/- 0.33Ω	+/- 2Ω

So what?

From the previous slide we can conclude for the 50ohm trace that the 2D solver gives a range of 50.02 +/- 2 ohms

Our measurements (**50.122 +/- 0.2 ohms**) **falls** within this range.

All of the field solvers and online calculator's results are within the uncertainty of our measurements.

Thank you

Any Questions?